**ELE226 Circuit Theory II Homework 2-3**

# 1 Questions

### Homework 2

**2.1.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Assessment Problem 12.2, p. 440*

**2.2.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.13, p. 460*

**2.3.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.20, p. 460*

**2.4.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.22, p. 460*

**2.5.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.24, p. 461*

**2.6.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.40, p. 462*

**2.7.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Assessment Problem 12.5, p. 447*

**2.8.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.41, p. 462*

**2.9.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.43, p. 463*

**2.10.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.43, p. 463*

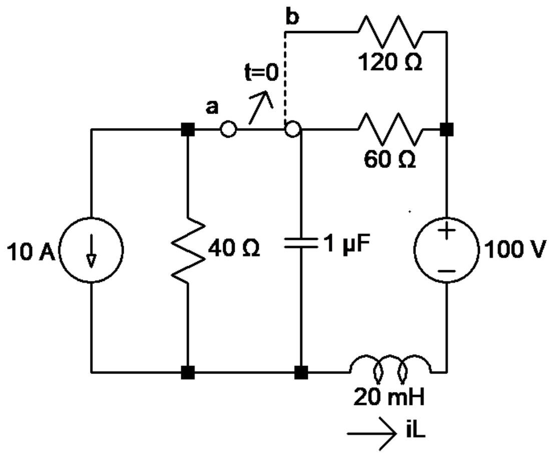
### Homework 3

**3.1.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.27, p. 461*

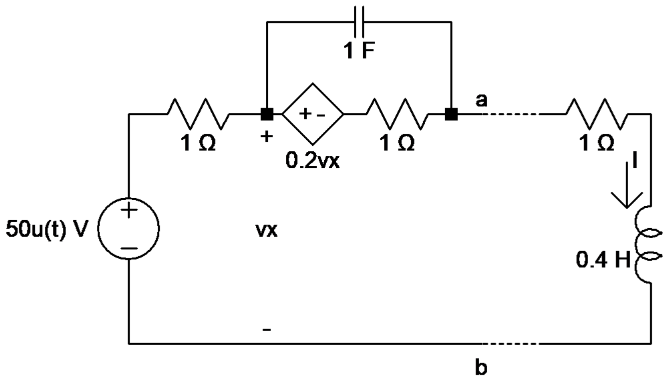
**3.2.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.28, p. 461*

**3.3.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.28, p. 461*

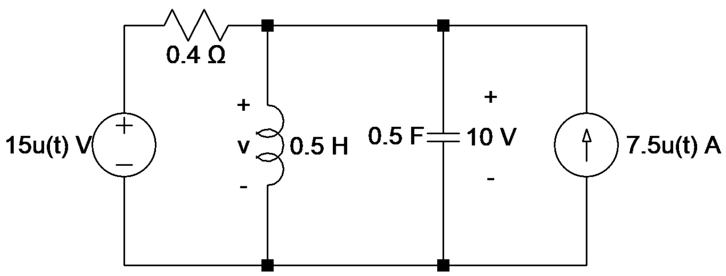
**3.4.** The switch has been in position $a$ for a long time. At $t = 0$, it moves to position $b$. Find $i_L(t)$ for $t \geq 0$. Verify $i_{L_{ss}}$ using the phasor approach.




**3.5.** The capacitor is initially uncharged. Find $I(s)$.




**3.6.** Find $v(t)$, $t > 0$ using the superposition principle.




**3.7.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.9, p. 505*

**3.8.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.11, p. 505*

**3.9.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.13, p. 506*

**3.10.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.14, p. 506*

**3.11.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.17, p. 506*

**3.12.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.20, p. 507*

**3.13.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.23, p. 507*

**3.14.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.28, p. 508*

**3.15.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.29, p. 508*

**3.16.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.30, p. 508*

**3.17.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.32, p. 509*

**3.18.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.35, p. 509*

**3.19.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.36, p. 509*

**3.20.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.36, p. 509*

## 2 Answers

### Homework 2

**2.1.** **a)** $\mathscr{L}\left\{t^2\right\} = \dfrac{2}{s^3} \implies \mathscr{L}\left\{t^2 e^{-at}\right\} = \boxed{\dfrac{2}{(s+a)^3}}.$

**b)**
$$\begin{aligned}\mathscr{L}\left\{\frac{d}{dt}(e^{-at}\sinh\beta t)\right\} &= s\mathscr{L}\left\{e^{-at}\sinh\beta t\right\} - e^{-at}\sinh\beta t|_{t=0} \\ &= s\cdot\frac{\beta}{(s+a)^2-\beta^2} - 1\cdot 0 = \boxed{\frac{s\beta}{(s+a)^2-\beta^2}}.\end{aligned}$$

**c)** $\mathscr{L}\left\{t\cos\omega t\right\} = -\dfrac{d}{ds}\left(\dfrac{s}{s^2+\omega^2}\right) = -\dfrac{s^2+\omega^2-s\cdot 2s}{(s^2+\omega^2)^2} = \boxed{\dfrac{s^2-\omega^2}{(s^2+\omega^2)^2}}.$

**2.2.** **a)** Remember: $\mathscr{L}\left\{g(t-a)u(t-a)\right\} = e^{-as}F(s)$. Let $g(t) = 20e^{-500t}$. Then

$$\mathscr{L}\{g(t)\} = \frac{20}{s+500} \implies \mathscr{L}\{g(t-a)u(t-a)\} = \boxed{\frac{20e^{-10s}}{s+500}}$$

**b)** Rearrange the equation:

$$f(t) = u(t+4)(5t+20) + u(t+2)(-10t-20) + u(t-2)(10t-20) + u(t-4)(5t-20)$$

$$\mathscr{L}\left\{f(t)\right\} = \boxed{5e^{4s}\cdot\frac{1}{s^2} - 10e^{2s}\cdot\frac{1}{s^2} + 10e^{-2s}\cdot\frac{1}{s^2} + 5e^{-4s}\cdot\frac{1}{s^2}}$$

**2.3.** **a)** $\mathscr{L}\left\{t\right\} = \dfrac{1}{s^2} \implies \mathscr{L}\left\{te^{-at}\right\} = \boxed{\dfrac{1}{(s+a)^2}}.$

**b)** $\mathscr{L}\{f(t)\} = \boxed{\dfrac{w}{s^2+\omega^2}}$

**c)** Use the formula $\sin(a+b) = \sin a\cos b + \cos a \sin b$.

$$\sin(\omega t+\theta) = \sin(\omega t)\cos\theta + \cos(\omega t)\sin\theta \implies \mathscr{L}\{f(t)\} = \boxed{\frac{\omega\cos\theta + s\sin\theta}{s^2+w^2}}$$

**d)** $\mathscr{L}\{f(t)\} = \boxed{\dfrac{1}{s^2}}$

**e)** Use the formula $\cosh(a+b) = \cosh a\cosh b + \sinh a\sinh b$.

$$\cosh(t+\theta) = \cosh t\cosh\theta + \sinh t\sinh\theta \implies \mathscr{L}\{f(t)\} = \boxed{\frac{s\cosh\theta - \sinh\theta}{s^2-1}}$$

**2.4.** **a)** $\mathscr{L}\left\{\dfrac{d}{dt}(\sin\omega t)\right\} = s\mathscr{L}\left\{\sin\omega t\right\} - \sin\omega t|_{t=0} = s\cdot\dfrac{\omega}{s^2+\omega^2} - 0 = \boxed{\dfrac{s\omega}{s^2+\omega^2}}.$

**b)** $\mathscr{L}\left\{\dfrac{d}{dt}(\cos\omega t)\right\} = s\mathscr{L}\left\{\cos\omega t\right\} - \cos\omega t|_{t=0} = s\cdot\dfrac{s}{s^2+\omega^2} - 1 = \boxed{\dfrac{-\omega^2}{s^2+\omega^2}}.$

**c)**
$$\begin{aligned}\mathscr{L}\left\{\frac{d^3}{dt^3}(t^2u(t))\right\} &= s^3\mathscr{L}\left\{t^2u(t)\right\} - s^2t^2u(t)|_{t=0} - s(t^2u(t))'|_{t=0} - (t^2u(t))''|_{t=0} \\ &= s^3\cdot\frac{2}{s^3} - 0 - s(2tu(t)+t^2\delta(t))|_{t=0} - (2u(t)+2t\delta(t))|_{t=0} \\ &= \boxed{2}.\end{aligned}$$

**d)** (a) $\rightarrow \mathscr{L}\{\omega\cos\omega t\} = \omega\mathscr{L}\{\cos\omega t\} = \boxed{\dfrac{s\omega}{s^2+\omega^2}}$.

(b) $\rightarrow \mathscr{L}\{-\omega\sin\omega t\} = -\omega\mathscr{L}\{\sin\omega t\} = \boxed{\dfrac{-\omega^2}{s^2+\omega^2}}$.

(c) $\rightarrow \mathscr{L}\{2\delta(t)\} = 2e^{-0s} = \boxed{2}$.

**2.5. a)**

$$\begin{aligned}\mathscr{L}\left\{\frac{d}{dt}(e^{-at}\cos\omega t)\right\} &= s\mathscr{L}\left\{e^{-at}\cos\omega t\right\} - e^{-at}\cos\omega t|_{t=0} \\ &= s\cdot\frac{s+a}{(s+a)^2+\omega^2} - 1 = \boxed{\frac{-as-a^2-\omega^2}{(s+a)^2+\omega^2}}.\end{aligned}$$

**b)** $\mathscr{L}\left\{\displaystyle\int_{0^-}^{t} e^{-ax}\sin\omega x\,dx)\right\} = \dfrac{\mathscr{L}\{e^{-ax}\sin\omega x\}}{s} = \boxed{\dfrac{w}{[(s+a)^2+\omega^2]s}}$.

**2.6. a)** $F(s) = \dfrac{K_1}{s+5} + \dfrac{K_2}{s+8}$.

$$K_1 = \left.\frac{6(s+10)}{s+8}\right|_{s=-5} = 10,\ K_2 = \left.\frac{6(s+10)}{s+5}\right|_{s=-8} = -4 \implies \boxed{f(t) = 10e^{-5t} - 4e^{-8t}}$$

**b)** $F(s) = \dfrac{K_1}{s} + \dfrac{K_2}{s+3} + \dfrac{K_2}{s+7}$.

$$K_1 = \left.\frac{20s^2+141s+315}{s^2+10s+21}\right|_{s=0} = 15,\ K_2 = \left.\frac{20s^2+141s+315}{s(s+3)}\right|_{s=-7} = 11$$

$$K_3 = \left.\frac{20s^2+141s+315}{s(s+7)}\right|_{s=-3} = 6 \implies \boxed{f(t) = 15 - 6e^{-3t} + 11e^{-7t}}$$

**c)** $F(s) = \dfrac{K_1}{s+2} + \dfrac{K_2}{s+4} + \dfrac{K_3}{s+6}$.

$$K_1 = \left.\frac{15s^2+112s+228}{(s+4)(s+6)}\right|_{s=-2} = 8,\ K_2 = \left.\frac{15s^2+112s+228}{(s+2)(s+6)}\right|_{s=-4} = -5,$$

$$K_3 = \left.\frac{15s^2+112s+228}{(s+2)(s+4)}\right|_{s=-6} = 12 \implies \boxed{f(t) = 8e^{-2t} - 5e^{-4t} + 12e^{-6t}}$$

**d)** $F(s) = \dfrac{K_1}{s} + \dfrac{K_2}{s+1} + \dfrac{K_3}{s+2} + \dfrac{K_4}{s+3}$.

$$K_1 = \left.\frac{2s^3+33s^2+93s+54}{(s+1)(s+2)(s+3)}\right|_{s=0} = 9,\ K_2 = \left.\frac{2s^3+33s^2+93s+54}{s(s+2)(s+3)}\right|_{s=0} = 4$$

$$K_3 = \left.\frac{2s^3+33s^2+93s+54}{s(s+1)(s+3)}\right|_{s=0} = -8,\ K_4 = \left.\frac{2s^3+33s^2+93s+54}{s(s+1)(s+2)}\right|_{s=0} = -3$$

$$\implies \boxed{f(t) = 9 + 4e^{-t} - 8e^{-2t} - 3e^{-3t}}$$

**2.7.** We have $F(s) = \dfrac{K_1}{s+5} + \dfrac{K_2}{s+5-j12} + \dfrac{K_2^*}{s+5+j12}$.

$$K_1 = \left.\frac{10(s^2+119)}{s^2+10s+169}\right|_{s=-5} = 10,\ K_2 = \left.\frac{10(s^2+119)}{(s+5)(s+5+j12)}\right|_{s=-5+j12} = j4.16$$

$$\implies \boxed{f(t) = 10e^{-5t} + 8.33e^{-5t}\cos(12t+90^\circ)}$$

**2.8.** **a)** $F(s) = \dfrac{K}{s+7-j14} + \dfrac{K^*}{s+7+j14}$.

$$K = \left.\frac{280}{s+7-j14}\right|_{s=-7-j14} = j10 \implies \boxed{f(t) = 20e^{-7t}\cos(14t - 90°)}$$

**b)** $F(s) = \dfrac{K_1}{s} + \dfrac{K_2}{s+5-j8} + \dfrac{K_2^*}{s+5+j8}$

$$K_1 = \left.\frac{-s^2+52s+445}{s^2+10s+89}\right|_{s=0} = 5,\ K_2 = \left.\frac{-s^2+52s+445}{s(s+5+j8)}\right|_{s=-5+j8} = -3+j2$$

$$\boxed{f(t) = 5 + 7.21e^{-5t}\cos(8t - 146.31°)}$$

**c)** $F(s) = \dfrac{K_1}{s+6} + \dfrac{K_2}{s+2-4j} + \dfrac{K_2^*}{s+2+4j}$.

$$K_1 = \left.\frac{14s^2+56s+152}{s^2+4s+20}\right|_{s=-6} = 10,\ K_2 = \left.\frac{14s^2+56s+152}{(s+2+j4)(s+6)}\right|_{s=-2+j4} = 2+j2$$

$$\implies \boxed{f(t) = 10e^{-6t} + 5.66e^{-2t}\cos(4t + 45°)}$$

**d)** $F(s) = \dfrac{K_1}{s+5-j3} + \dfrac{K_1^*}{s+5+j3} + \dfrac{K_2}{s+4-j2} + \dfrac{K_2^*}{s+4+j2}$

$$K_1 = \left.\frac{8(s+1)^2}{(s+5+j3)(s^2+8s+20)}\right|_{s=-5+j3} = 3.54 - j2.97$$

$$K_2 = \left.\frac{8(s+1)^2}{(s^2+10s+34)(s+4+2j)}\right|_{s=-4+j2} = -3.54 + j0.69$$

$$\boxed{f(t) = 9.25e^{-5t}\cos(3t - 40.04°) + 7.22e^{-4t}\cos(2t + 168.93°)}$$

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**2.9.** **a)** $F(s) = \dfrac{K_1}{s^2} + \dfrac{K_2}{s} + \dfrac{K_3}{s+8}$.

$$K_1 = \left.\frac{320}{s+8}\right|_{s=0} = 40,\ K_2 = \left.\frac{d}{ds}\left(\frac{320}{s+8}\right)\right|_{s=0} = -5,\ K_3 = \left.\frac{320}{s^2}\right|_{s=-8} = 5$$

$$\boxed{f(t) = 40t - 5 + 5e^{-8t}}$$

**b)** $F(s) = \dfrac{K_1}{s} + \dfrac{K_2}{(s+2)^2} + \dfrac{K_3}{s+2}$.

$$K_1 = \left.\frac{80(s+3)}{(s+2)^2}\right|_{s=0} = 60,\ K_2 = \left.\frac{80(s+3)}{s}\right|_{s=-2} = -40$$

$$K_3 = \frac{d}{ds}\left[\frac{80(s+3)}{s}\right]_{s=-2} = -60 \implies \boxed{f(t) = 60 - 40te^{-2t} - 60e^{-2t}}$$

**c)** $F(s) = \dfrac{K_1}{(s+1)^2} + \dfrac{K_2}{s+1} + \dfrac{K_3}{s+3-j4} + \dfrac{K_4}{s+3+j4}.$

$$K_1 = \left.\frac{60(s+5)}{s^2+6s+25}\right|_{s=-1} = 12,\ K_2 = \frac{d}{ds}\left[\frac{60(s+5)}{s^2+6s+25}\right]_{s=-1} = 0.6$$

$$K_3 = \frac{d}{ds}\left[\frac{60(s+5)}{(s+1)^2(s+3+j4)}\right]_{s=-3+j4} = -0.3 + j1.65$$

$$\implies \boxed{f(t) = 12te^{-t} + 0.6e^{-t} + 3.35e^{-3t}\cos(4t + 100.305°)}$$

**d)** $F(s) = \dfrac{K_1}{(s+5)^2} + \dfrac{K_2}{s+5} + \dfrac{K_3}{s^2} + \dfrac{K_4}{s}.$

$$K_1 = \left.\frac{25(s+4)^2}{s^2}\right|_{s=-5} = 1,\ K_2 = \frac{d}{ds}\left[\frac{25(s+4)^2}{s^2}\right]_{s=-5} = -1.6$$

$$K_3 = \left.\frac{25(s+4)^2}{(s+5)^2}\right|_{s=-5} = 16,\ K_4 = \frac{d}{ds}\left[\frac{25(s+4)^2}{(s+5)^2}\right]_{s=-5} = 1.6$$

$$\implies \boxed{f(t) = 1.6 + 16t + te^{-5t} - 1.6e^{-5t}}$$

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**2.10.** **a)** $F(s) = \dfrac{K_1}{s} + \dfrac{K_2}{(s+3)^3} + \dfrac{K_3}{(s+3)^2} + \dfrac{K_4}{s+3}.$

$$K_1 = \left.\frac{135}{(s+3)^3}\right|_{s=0} = 5,\ K_2 = \left.\frac{135}{s}\right|_{s=-3} = -45$$

$$K_3 = \left.\frac{d}{ds}\left(\frac{135}{s}\right)\right|_{s=-3} = -15,\ 2K_4 = \left.\frac{d^2}{ds^2}\left(\frac{135}{s}\right)\right|_{s=-3} = -10. \implies K_4 = -5$$

$$\boxed{f(t) = 5 - 5e^{-3t} - 15te^{-3t} - 22.5t^2e^{-3t}}$$

**b)** $F(s) = \dfrac{As+B}{(s^2+2s+2)^2} + \dfrac{Cs+D}{s^2+2s+2}.$

$$10(s+2)^2|_{s=-1+j} = -j20 = A(-j+1) + B \implies A = 20,\ B = 20$$

$$\left.\frac{d}{ds}[10(s+2)]^2\right|_{s=-1+j} = j20 + 20 = A + C(s^2+2s+2) + (Cs+D)(2s+2)|_{s=-1+j}$$

$$\implies C = 0,\ D = 10.$$

$$\boxed{f(t) = 10(t+1)e^{-t}\sin t}$$

**c)** $F(s) = 25 + \dfrac{20s+114}{s^2+15s+54} = 25 + \dfrac{K_1}{s+9} + \dfrac{K_1}{s+6}$

$$K_1 = \left.\frac{20s+144}{s+6}\right|_{s=-9} = 12,\ K_2 = \left.\frac{20s+144}{s+9}\right|_{s=-6} = 8$$

$$\implies \boxed{f(t) = 25\delta(t) + 8e^{-6t} + 12e^{-9t}}$$

**d)** $F(s) = 5s - 15 + \dfrac{K_1}{s+2} + \dfrac{K_2}{s+5}$

$$K_1 = \left.\frac{6s+42}{s+5}\right|_{s=-2} = 10,\ K_2 = \left.\frac{6s+42}{s+2}\right|_{s=-5} = -4$$

$$\implies \boxed{f(t) = 5\delta'(t) - 15\delta(t) + 10e^{-2t} - 4e^{-5t}}$$

**Homework 3**

**3.1. a.** $\boxed{-I_{\text{dc}} + \dfrac{1}{L}\displaystyle\int_0^t v_o(\tau)\,d\tau + \dfrac{v_o}{R} + C\dfrac{dv_o}{dt} = 0}$

**b.** $\dfrac{-I_{\text{dc}}}{s} + \dfrac{1}{L}\cdot\dfrac{V_o}{s} + \dfrac{V_o}{R} + CsV_o + \cancel{CV_o(0)} = 0 \implies \boxed{V_o(s) = \dfrac{I_{\text{dc}}/C}{s^2 + \frac{s}{RC} + \frac{1}{LC}} \text{ V–s}}$

**c.** $i_o = C\dfrac{dv_o}{dt} \implies \boxed{I_o(s) = sCV_o(s) = \dfrac{sI_{\text{dc}}}{s^2 + \frac{s}{RC} + \frac{1}{LC}} \text{ A–s}}$

**3.2. a.** $\boxed{\dfrac{v_o}{R} + C\dfrac{dv_o}{dt} + \dfrac{1}{L}\displaystyle\int_0^t v_o(\tau)\,d\tau = 0}$

**b.** We have $\dfrac{L}{R}\dfrac{di_o}{dt} + LC\dfrac{d^2i_o}{dt^2} + i_o = 0.$

$$\frac{L}{R}sI_o - \frac{L}{R}I_{\text{dc}} + LC(s^2I_o - sI_{\text{dc}} - 0) + I_o = 0 \implies I_o\left(\frac{sL}{R} + LCs^2 + 1\right) - I_{\text{dc}}\left(\frac{L}{R} + sLC\right)$$

$$\implies \boxed{I_o(s) = \frac{I_{\text{dc}}\left(s + \frac{1}{RC}\right)}{s^2 + \frac{s}{RC} + \frac{1}{LC}} \text{ A–s}}$$

**3.3. a.** Let $i_o$ circulate in the clockwise direction.

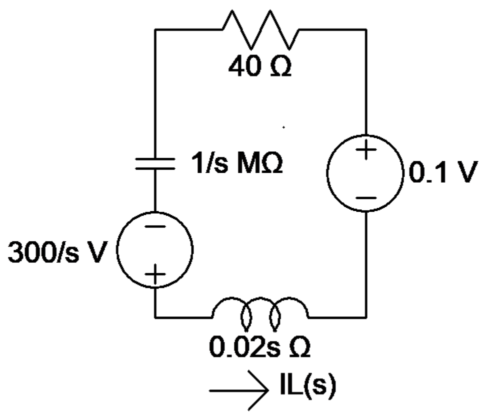
$$Ri_o + L\frac{di_o}{dt} + v_o = 0 \ \overset{\left[i_o = C\frac{dv_o}{dt}\right]}{\implies} \ \boxed{\frac{d^2v_o}{dt^2} + \frac{R}{L}\frac{dv_o}{dt} + \frac{1}{LC}v_o = 0}$$

**b.** From KVL,

$$s^2V_o - sV_{\text{dc}} - \cancel{v_o'(0)} + \frac{R}{L}(sV_o - V_{\text{dc}}) + \frac{1}{LC}V_o = 0 \implies \boxed{V_o(s) = \frac{V_{\text{dc}}\left(s + \frac{R}{L}\right)}{s^2 + \frac{R}{L}s + \frac{1}{LC}} \text{ V–s}}$$

**3.4.** Initial conditions:

$$i_L(0^-) = 5 \text{ A}, \quad v_C(0^-) = -200 \text{ V}.$$



From KVL,

$$-0.1-\frac{300}{s}-40I_L+\frac{10^6}{s}I_L+20sI_L\cdot 10^{-3}=0 \implies I_L=\frac{0.1+\dfrac{300}{s}}{\dfrac{10^6}{s}+40+0.02s}$$

$$I_L=\frac{5s+15000}{s^2+2000s+50\cdot 10^6}=\frac{K_1}{s+1000-j7000}+\frac{K_1^*}{s+1000+j7000}$$

$$K_1=\left.\frac{5s+15000}{s+1000+j7000}\right|_{s=-1000+j7000} \qquad z=2.5-j0.714$$

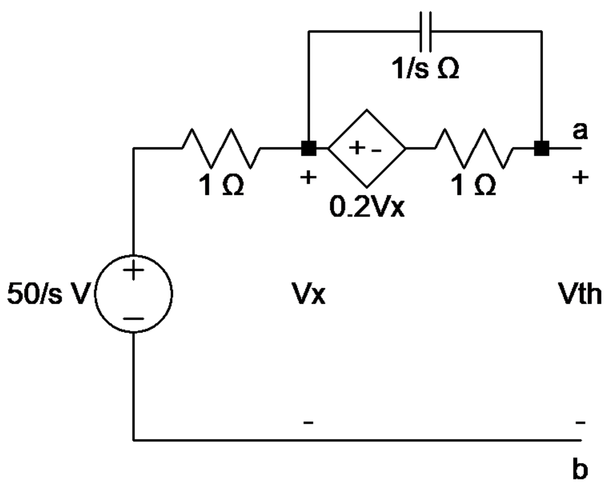
$$\implies \boxed{i_L(t)=[5.2e^{-1000t}\cos(7000t-15.945°)]u(t)\text{ A}}$$

The steady part is 0 because the capacitor is open at infinity $\left(\omega=0 \implies \frac{1}{j\omega C}\to\infty\right)$, leading to $i_{L_{ss}}=0$.

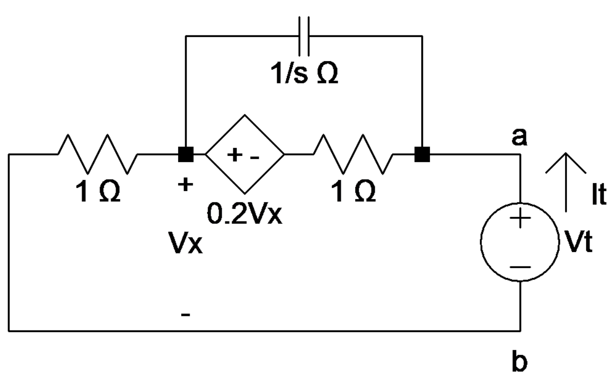
**3.5.** Calculate $\text{V}_{Th}$.

$$\frac{V_x-\dfrac{50}{s}}{1}+\frac{V_x-V_{Th}}{1/s}+\frac{V_x-0.2V_x-V_{Th}}{1}=0 \quad \text{and} \quad V_x=\frac{50}{s}$$

$$\implies V_{th}=\frac{50s+40}{s+s^2}$$




Zero the 50 V source and apply a test voltage.

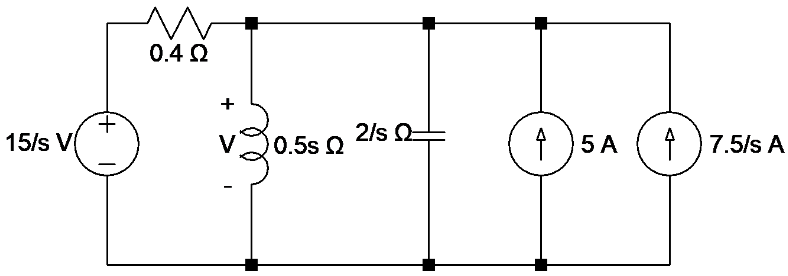


$$\frac{V_x}{1}+\frac{V_x-0.2V_x-V_t}{1}+\frac{V_x-V_t}{1/s}=0 \quad \text{and} \quad V_x=i_t$$

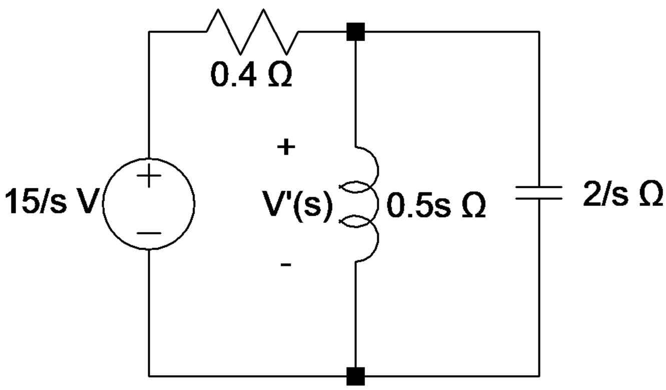
$$I_t(2-0.2+s)=V_t(s+1) \implies \frac{V_t}{I_t}=R_{Th}=\frac{1.8+s}{s+1}$$

$$\boxed{I(s)=\frac{\dfrac{50s+40}{s(s+1)}}{1+0.4s+\dfrac{s+1.8}{s+1}}=\frac{125s+100}{s(s^2+6s+7)} \text{ A–s}}$$

**3.6.** The $s$–domain equivalent circuit:



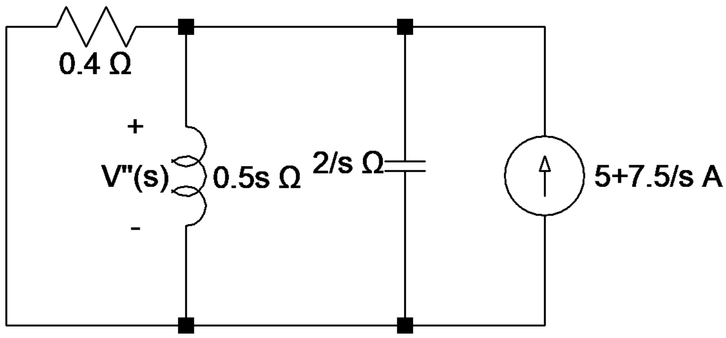

Zero the current source.




$$V'(s)=\frac{15}{s}\cdot\frac{0.5s \parallel \dfrac{2}{s}}{\left(0.5s \parallel \dfrac{2}{s}\right)+0.4}=\frac{75}{(s+1)(s+4)}=\frac{K_1}{s+1}+\frac{K_2}{s+4}$$

$$K_1=\left.\frac{75}{s+4}\right|_{s=-1}=25, \quad K_2=\left.\frac{75}{s+1}\right|_{s=-4}=-25 \implies V'(s)=\frac{25}{s+1}-\frac{25}{s+4}$$

Zero the voltage source.

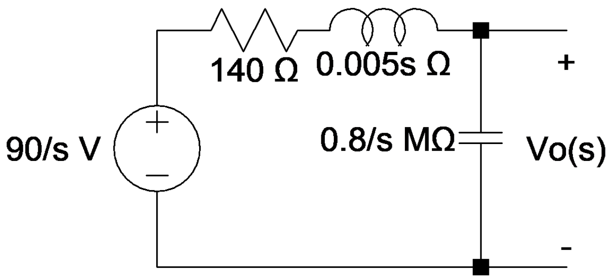



$$V''(s) = \left(\frac{7.5}{s} \parallel s\right) \parallel \left(\frac{2}{s} \parallel 0.5s \parallel 0.4\right) = \frac{2s}{s^2+5s+4} = \frac{K_1}{s+4} + \frac{K_2}{s+1}$$

$$K_1 = \left.\frac{15+10s}{s+1}\right|_{s=-4} = \frac{25}{3}, \quad K_2 = \left.\frac{15s+10}{s+4}\right|_{s=-1} = \frac{5}{3} \implies V''(s) = \frac{25}{3(s+4)} + \frac{5}{3(s+1)}$$

$$V(s) = V'(s) + V''(s) = -\frac{50}{3}\frac{1}{s+4} + \frac{80}{3}\frac{1}{s+1} \implies \boxed{v(t) = \left(-\frac{50}{3}e^{-4t} + \frac{80}{3}e^{-t}\right)u(t)\ \mathrm{V}}$$

**3.7.** Find the $s$–domain circuit.



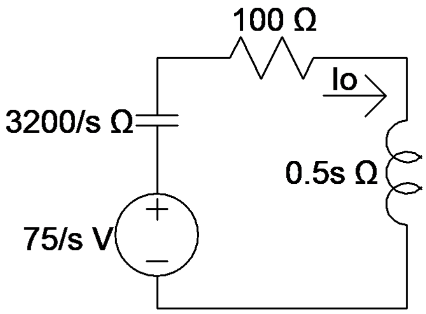

$$\begin{aligned} V_o(s) &= \frac{90}{s} \cdot \frac{800000/s}{\frac{800000}{s} + 0.005s + 140} = \frac{90}{s} \cdot \frac{16 \cdot 10^7}{s^2 + 28000s + 16 \cdot 10^7} \\ &= \frac{144 \cdot 10^8}{s(s+20000)(s+8000)} = \frac{K_1}{s} + \frac{K_2}{s+20000} + \frac{K_3}{s+8000} \end{aligned}$$

$$K_1 = \left.\frac{144 \cdot 10^9}{(s+2000)(s+8000)}\right|_{s=0} = 90, \ K_2 = \left.\frac{144 \cdot 10^9}{s(s+8000)}\right|_{s=-20000} = 60$$

$$K_3 = \left.\frac{144 \cdot 10^9}{s(s+20000)}\right|_{s=-8000} = -150$$

$$\implies \boxed{v_o(t) = [90 - 150e^{-8000t} + 60e^{-20000t}]u(t)\ \mathrm{V}}$$

**3.8.** **a)** The circuit:




**b)** $-\dfrac{75}{s} + I_o\dfrac{3200}{s} + 100I_0 + 0.5sI_o = 0 \implies \boxed{I_o(s) = \dfrac{150}{(s+40)(s+160)} \text{ A–s}}$

**c)** $I_o(s) = \dfrac{K_1}{s+40} + \dfrac{K_2}{s+160}$

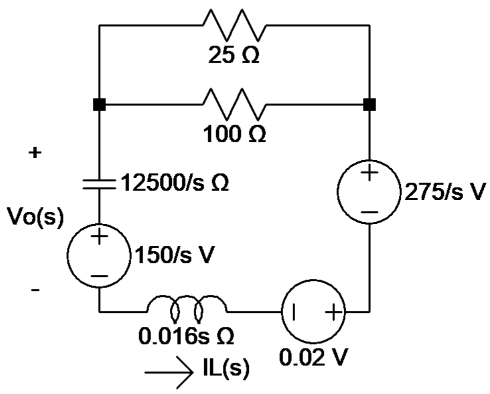
$$K_1 = \left.\frac{150}{s+160}\right|_{s=40} = \frac{5}{4}, \quad K_2 = \left.\frac{150}{s+40}\right|_{s=-160} = -\frac{5}{4}$$

$$\implies \boxed{i_o(t) = \frac{5}{4}\left(e^{-40t} - e^{-160t}\right)u(t) \text{ A}}$$

**3.9.** Find the current through the inductor.

$$-1.25 + \frac{v_C}{80} + \frac{v_C}{200} + \frac{v_C - 275}{100} = 0 \implies v_C = 150 \text{ V}, \qquad i_L = \frac{275-150}{100} = 1.25 \text{ A}$$

**a)** The circuit:




**b)** From KCL,

$$0.016sI_L - 0.02 - \frac{2.75}{s} + 20I_L + \frac{12500}{s}I_L + \frac{150}{s} = 0 \implies I_L = \frac{7812.5 + 1.25s}{s^2 + 1250s + 781250}$$

$$\boxed{V_o = \frac{12500}{s}I + \frac{150}{s} = \frac{150s^2 + 203125s + 214843750}{s(s^2 + 1250s + 781250)} \text{ V–s}}$$

**c)** From (b)

$$V_o = \frac{K_1}{s} + \frac{K_2}{s+625-j625} + \frac{K_2^*}{s+625+j625}$$

$$K_1 = \left.\frac{150s^2 + 203125 + 214843750}{s^2 + 1250s + 781250}\right|_{s=0} = 275$$

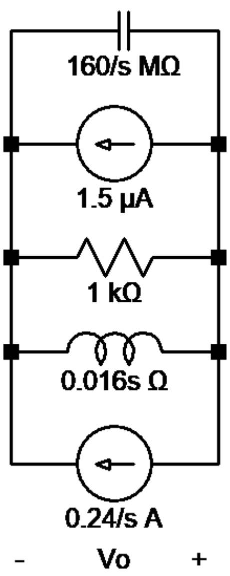
$$K_2 = \left.\frac{150s^2 + 203125 + 214843750}{s(s + 625 + j625)}\right|_{s=-625+j625} = 80.04\underline{/141.34^\circ}$$

$$\implies \boxed{v_o(t) = [275 + 160.08e^{-625t}\cos(625t + 141.34^\circ)]u(t)\ \text{V}}$$

**3.10.** Initial conditions:

$$v_c(0^-) = v_c(0) = 360 \cdot \frac{4\,k \parallel 1\,k}{4\,k \parallel 1\,k + 400} = 240\ \text{V}, \quad i_L(0^-) = i_L(0) = \frac{240}{1\,k} = 0.24\ \text{A}$$

**a)** The circuit:




**b)** $Z_{eq} = \dfrac{16 \cdot 10^7}{s} \parallel 1000 \parallel 0.016s, \quad I_{eq} = 1.5 \cdot 10^{-6} + \dfrac{0.24}{s}$

$$V_o = -I_{eq}Z_{eq} = -\frac{1.5 \cdot 10^{-6} + \frac{0.24}{s}}{\frac{1}{\frac{16\cdot 10^7}{s}} + \frac{1}{1000} + \frac{1}{0.016s}} = \boxed{-\frac{240s + 38.4 \cdot 10^6}{s^2 + 160000s + 10^{10}}\ \text{V–s}}$$

**c)** $I_L = \dfrac{V_o}{0.016s} + \dfrac{0.24}{s} = \boxed{\dfrac{0.24s + 23400}{s^2 + 160000s + 10^{10}}\ \text{A–s}}$

**d)** From (b), we have

$$V_o = -\frac{240s + 38.4 \cdot 10^6}{s^2 + 160000s + 10^{10}} = \frac{K_1}{s + 80000 - j60000} + \frac{K_1^*}{s + 80000 + j60000}$$

$$K_1 = \left.\frac{-240s - 38400000}{s + 80000 + j60000}\right|_{s=-80000+j60000} = 200\underline{/126.87^\circ}$$

$$\implies \boxed{v_o(t) = [400e^{-80000t}\cos(60000t + 126.87^\circ)]u(t)\ \text{A}}$$

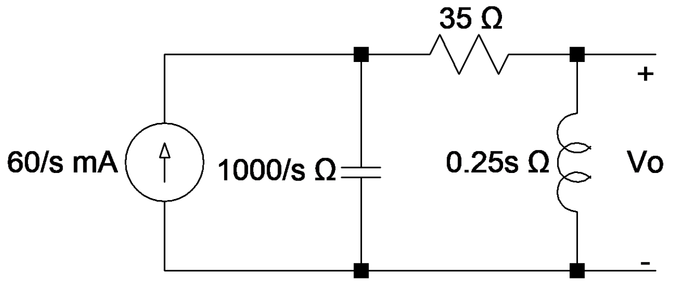
**e)** From (b), we have

$$I_o = \frac{0.24s + 23400}{s^2 + 160000s + 10^{10}} = \frac{K_1}{s + 80000 - j60000} + \frac{K_1^*}{s + 80000 + j60000}$$

$$K_1 = \left.\frac{0.24s + 23400}{s + 80000 + j60000}\right|_{s=-80000+j60000} = 0.125\underline{/-16.26^\circ}$$

$$\implies \boxed{i_L(t) = [0.25e^{-80000t}\cos(60000t - 16.26^\circ)]u(t)\ \text{A}}$$

**3.11.** The circuit in the $s$–domain:




**a)** $V_o = \dfrac{60}{s^2} \cdot \dfrac{0.25s}{\dfrac{1600}{s} + 35 + 0.25s} = \dfrac{60}{s^2 + 140s + 4000}$

**b)** $\lim\limits_{t\to 0} v_o(t) = \lim\limits_{s\to\infty} sV_o(s) = \dfrac{s}{s^2 + 140s + 4000} = 0$

$\lim\limits_{t\to\infty} v_o(t) = \lim\limits_{s\to 0} sV_o(s) = \dfrac{s}{s^2 + 140s + 4000} = 0.$

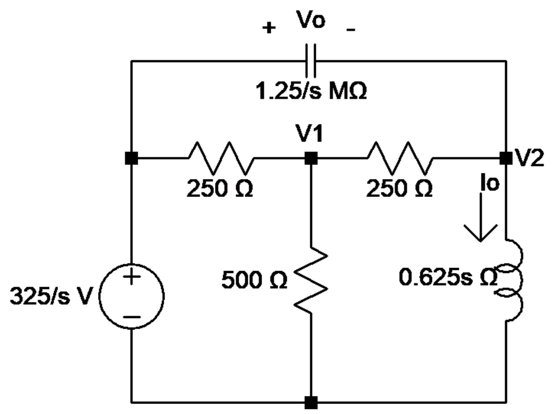
$$\boxed{v_o(0) = v_o(\infty) = 0}$$

**c)** From (a), we have

$$V_o = \frac{60}{s^2 + 140s + 4000} = \frac{K_1}{s + 100} + \frac{K_2}{s + 40} \quad [K_1 = -1,\ K_2 = 1 \text{ directly}]$$

$$\boxed{v_o(t) = (e^{-40t} - e^{-100t})u(t)\ \text{V}}$$

**3.12.** The circuit in the $s$–domain:




**a)** From KCL equations,

$$\frac{V_1 - \frac{325}{s}}{250} + \frac{V_1}{500} + \frac{V_1 - V_2}{250} = 0,\ \frac{V_2}{0.625s} + \frac{V_2 - V_1}{250} + \frac{V_2 - \frac{325}{s}}{1250000s} = 0 \implies 5V_1 - 2V_2 = \frac{650}{s}$$

$$-5000sV_1 + (s^2 + 5000s + 2000000)V_2 = 325s \implies V_2 = \frac{325}{s + 1000}$$

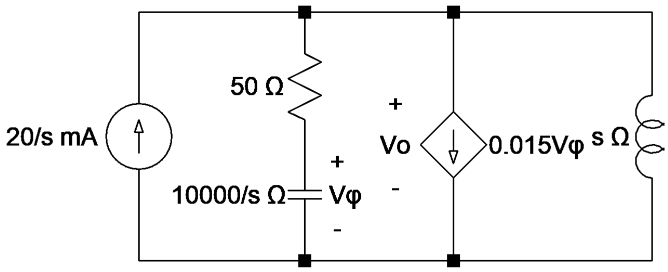
$$\boxed{\begin{aligned} V_o &= \frac{325}{s} - \frac{325}{s+1000} \text{ V–s} \\ I_o &= \frac{0.52}{s} - \frac{0.52}{s+1000} \text{ A–s} \end{aligned}}$$

**b)** Using the results in $(a)$,

$$\boxed{\begin{aligned} v_o(t) &= (325 - 325e^{-1000t})u(t) \text{ V} \\ i_o(t) &= (520 - 520e^{-1000t})u(t) \text{ mA} \end{aligned}}$$

**c)** The poles of functions of $s$ lie on the $y$–axis and on the left of the axis, leading to constant and decaying exponential functions of $t$, which is feasible.

**3.13.** The circuit in the $s$–domain:




From KCL,

$$\frac{0.02}{s} + \frac{V_o}{50 + 10^4/s} + 0.015V_\phi + \frac{V_o}{s} = 0.$$

We have $V_\phi = \dfrac{\frac{10^4}{s}}{50 + \frac{10^4}{s}} V_o = \dfrac{10^4 \cdot V_o}{50s + 10^4}$

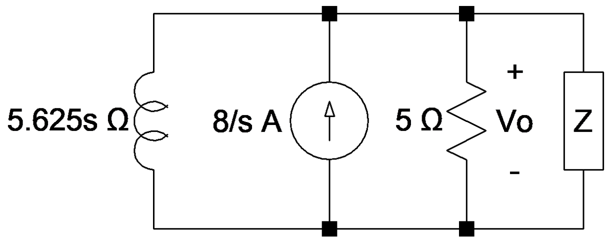
Replacing $V_\phi$ in the KCL equation and solving for $V_o$, we get

$$V_o = \frac{s+200}{s^2 + 200s + 10^4} = \frac{K_1}{(s+100)^2} + \frac{K_2}{s+100}$$

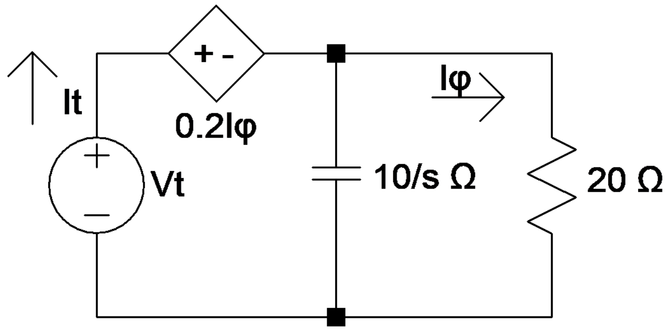
$$K_1 = (s+200)\big|_{s=-100} = 100, \quad K_2 = 1$$

$$\Longrightarrow \boxed{v_o(t) = (100te^{-100t} + e^{-100t})u(t) \text{ V}}$$

**3.14.** The $s$–domain equivalent circuit:




**a)** Apply a test voltage to find the impedance to the right.




$$V_t = 25I_\phi + \frac{20 \cdot \frac{10}{s}}{20 + \frac{10}{s}} I_t, \quad I_\phi = \frac{1 + \frac{10}{s}}{20 + \frac{10}{s}} \implies Z = \frac{V_t}{I_t} = \frac{250 + 200}{20s + 10} = \frac{45}{2s + 1}$$

$$\frac{V_o}{s} + \frac{V_o(2s+1)}{45} + \frac{V_o}{5.625s} = \frac{8}{s} \implies V_o = \frac{180}{s^2 + 5s + 4} \text{ V–s}$$

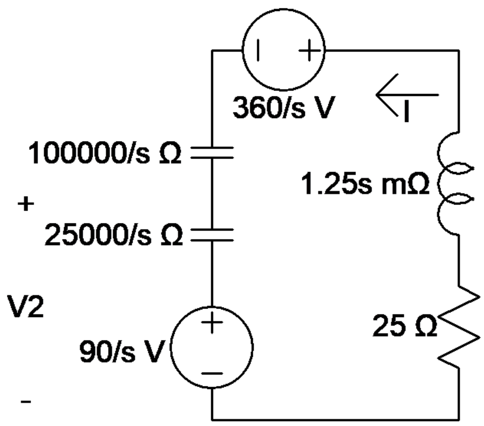
$$V_1 = \left.\frac{180}{s+1}\right|_{s=-4} = -60, \quad V_2 = \left.\frac{180}{s+4}\right|_{s=-1} = 60 \implies \boxed{V_o = \frac{60}{s+1} - \frac{60}{s+4} \text{ V–s}}$$

**b)** $\boxed{v_o(t) = (60e^{-t} - 60e^{-4t})u(t) \text{ V}}$

**3.15.** Initially, we have

$$v_2(0^-) = 450 \cdot \frac{24\,\mu \parallel 16\,\mu}{24\,\mu \parallel 16\,\mu + 10\,\mu} = 90 \text{ V}$$

**a)** The circuit:




**b)** Let $I$ be the current in the counterclockwise direction.

$$25I + 1.25 \cdot 10^{-3} sI + \frac{450}{s} = 0 \implies I = \frac{-450s}{25 + 1.25 \cdot 10^{-3}s + \frac{125000}{s}}$$

$$\boxed{V_1 = -25I = \frac{9000000}{(s+10000)^2} \text{ V–s}}$$

$$K_1 = 90s + 1800000|_{s=-10000} = 900000, \quad K_2 = 90$$

$$\implies \boxed{v_1(t) = (9000000te^{-10000t})u(t) \text{ V}}$$

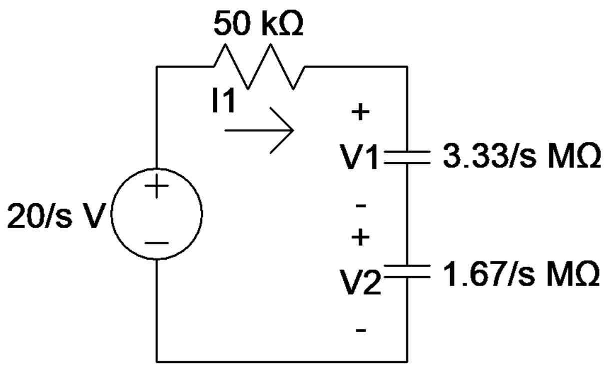
**c)** $v_2 = I \cdot (25000/s) + 90/s = \dfrac{90s + 1800000}{s^2 + 20000s + 10^8} = \dfrac{K_1}{(s+10000)^2} + \dfrac{K_2}{s+10000}$

$$K_1 = 90s + 18 \cdot 10^5|_{s=-10000} = 9 \cdot 10^5, \quad K_2 = 90$$

$$\boxed{V_2 = \frac{900000}{(s+10000)^2} + \frac{90}{s+10000} \text{ V–s}}$$

$$\implies \boxed{v_2(t) = [900000te^{-10000t} + 90e^{-10000t}]u(t) \text{ V}}$$

**3.16.** **a)** The circuit:




**b)** From KVL,

$$-\frac{20}{s} + 50000I_1 + \frac{5 \cdot 10^6 I_1}{s} = 0 \implies \boxed{I_1 = \frac{20}{5000000 + 50000s} = \frac{0.4 \cdot 10^{-3}}{s+100} \text{ A–s}}$$

$$\boxed{V_1 = \frac{10}{3} \cdot \frac{10^6}{s} \cdot I = \frac{4000/3}{s(s+100)} \text{ V–s}, \quad V_2 = \frac{10^7}{6s} \cdot I = \frac{2000/3}{s(s+100)} \text{ V–s}}$$

**c)** $\boxed{I = (0.4 \cdot 10^{-3} e^{-100t})u(t) \text{ A}}$

$$V_1 = \frac{K_1}{s} + \frac{K_2}{s+100} \implies K_1 = \left.\frac{4000/3}{s+100}\right|_{s=0} = \frac{40}{3},\ K_2 = \left.\frac{4000/3}{s}\right|_{s=-100} = -\frac{40}{3}$$

$$\boxed{V_1 = \frac{40}{3}(1 - e^{-100t})u(t) \text{ V}, \quad V_2 = \frac{V_1}{2} = \frac{20}{3}(1 - e^{-100t})u(t) \text{ V}}$$

**d)** The poles of functions of $s$ lie on the $y$–axis and on the left of the axis, leading to constant and decaying exponential functions of $t$, which is physically meaningful.

**3.17.** **a)** The Laplace transform of $V_m \sin(\omega t + \phi)$:

$$\mathscr{L}\{V_m \sin(\omega t + \phi)\} = V_m \mathscr{L}\{\sin \omega t \cos \phi + \cos \omega t \sin \phi\} = \frac{\omega \cos \phi + s \sin \phi}{s^2 + \omega^2}$$

$$I(s) = \frac{V_m \dfrac{\omega \cos \phi + s \sin \phi}{s^2 + \omega^2}}{sL + R} = \frac{\frac{V_m}{L}(\omega \cos \phi + s \sin \phi)}{(s + \frac{R}{L})(s^2 + \omega^2)} = \frac{K_1}{s + \frac{R}{L}} + \frac{K_2}{s - j\omega} + \frac{K_2^*}{s + j\omega}$$

$$K_1 = \left.\frac{\frac{V_m}{L}(\omega\cos\phi + s\sin\phi)}{s^2+\omega^2}\right|_{s=-\frac{R}{L}} = \frac{V_m(L\omega\cos\phi - R\sin\phi)}{R^2+\omega^2L^2}$$

$$K_2 = \left.\frac{\frac{V_m}{L}(\omega\cos\phi + s\sin\phi)}{\left(s+\frac{R}{L}\right)(s+j\omega)}\right|_{s=jw} = \frac{\frac{V_m}{L}(\omega\cos\phi + j\omega\sin\phi)}{\left(j\omega+\frac{R}{L}\right)2j\omega} = \frac{V_m(\cos\phi + j\sin\phi)}{(j\omega L+R)2j}$$

$$K_2 = \frac{V_m e^{j\phi}}{2\underline{/90^\circ}\sqrt{\omega^2L^2+R^2}\underline{/\tan^{-1}(\omega L/R)}} = \frac{V_m}{2\sqrt{\omega^2L^2+R^2}}\underline{/\phi - 90^\circ - \theta(\omega)}$$

$$\implies \boxed{i(t) = \underbrace{\frac{V_m(L\omega\cos\phi - R\sin\phi)}{\omega^2L^2+R^2}e^{-\frac{R}{L}t}}_{\text{Transient State}} + \underbrace{\frac{V_m\sin[\omega t+\phi-\theta(\omega)]}{\sqrt{\omega^2L^2+R^2}}}_{\text{Steady State}}\ \text{A}}$$

**b)** $\boxed{\frac{V_m(L\omega\cos\phi - R\sin\phi)}{\omega^2L^2+R^2}e^{-\frac{R}{L}t}\ \text{A}}$

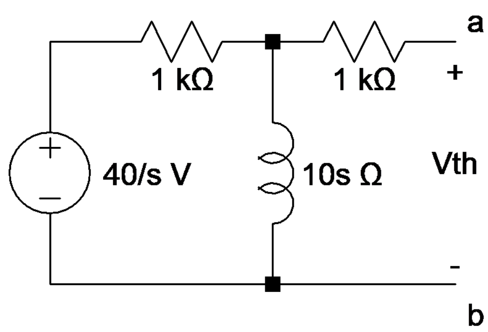
**c)** $\boxed{\frac{V_m\sin[\omega t+\phi-\theta(\omega)]}{\sqrt{\omega^2L^2+R^2}}\ \text{A}}$

**d)** $\mathbf{V}_g = \mathbf{I}(j\omega L+R) \implies \mathbf{I} = \frac{V_m\underline{/\phi - 90^\circ}}{\sqrt{\omega^2L^2+R^2}\underline{/\theta(\omega)}}$

$$\implies \boxed{\Re(\mathbf{I}) = \frac{V_m}{\sqrt{\omega^2L^2+R^2}}\sin[\omega t+\phi-\theta(\omega)]\ \text{A}}$$

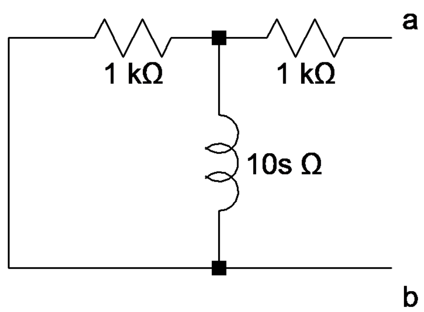
**e)** $L\omega\cos\phi = R\sin\phi \implies \boxed{\phi = \tan^{-1}\left(\frac{\omega L}{R}\right)}$

**3.18.** Calculate $V_{th}$.




$$V_{th} = \frac{40}{s}\cdot\frac{10s}{10s+1000} = \frac{40}{s+100}\ \text{V–s}$$

Calculate $Z_{th}$.



$$Z_{th} = 1000 + \frac{1}{\frac{1}{1000} + \frac{1}{10s}} = \frac{2000s + 100000}{s + 100}$$

$$I = \frac{\frac{40}{s+100}}{\frac{500000}{s} + \frac{2000s+1000}{s+100}} = \frac{40s}{2000s^2 + 600000s + 5000000} = \frac{0.02s}{s^2 + 30s + 25000}$$

$$I = \frac{K_1}{s + 150 - j50} + \frac{K_1^*}{s + 150 + j50}$$

$$\implies K_1 = \left.\frac{0.02s}{s + 150 + j50}\right|_{s=-150+j50} = 31.62 \cdot 10^{-3}\underline{/71.57^\circ}$$

$$\implies \boxed{i(t) = [63.24e^{-150t}\cos(50t + 71.57^\circ)]u(t)\text{ mA}}$$

**3.19.** **a)** Let $I_1$ and $I_2$ circulate clockwise in the left and right meshes, respectively.

$$(100 + 0.1875s)I_1 + 0.125sI_2 - \frac{150}{s} = 0 \quad \text{and} \quad 0.125sI_1 + (200 + 0.25s)I_1 = 0$$

Solve the latter equation for $I_1$ and substitute into the former equation, then

$$\implies \boxed{I_1(s) = \frac{1200(s + 800)}{s(s^2 + 2000s + 640000)}\text{ A–s}}$$

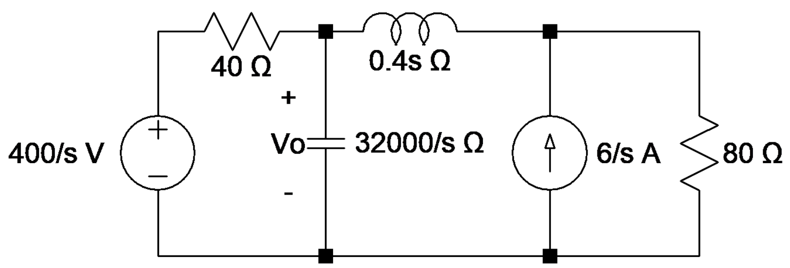
**b)** $\boxed{i_1(0) = \lim_{s\to\infty} sI_1 = 0, \quad \lim_{t\to\infty} i_1(t) = \lim_{s\to 0} sI_1 = 1.5\text{ A}}$

**c)** $I_1 = \frac{K_1}{s} + \frac{K_2}{s + 400} + \frac{K_3}{s + 1600}$

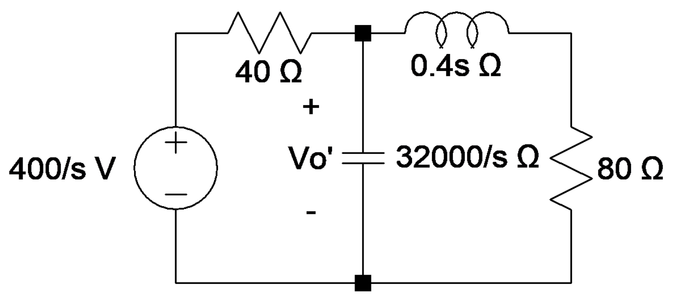
$$K_1 = \left.\frac{1200(s + 800)}{(s^2 + 1000s + 640000)}\right|_{s=0} = 1.5, \quad K_2 = \left.\frac{1200(s + 800)}{s(s + 1600)}\right|_{s=-400} = -1,$$

$$K_3 = \left.\frac{1200(s + 800)}{s(s + 400)}\right|_{s=-1600} = -0.5 \implies \boxed{i_1(t) = (1.5 - e^{-400t} - 0.5e^{-1600t})u(t)\text{ A}}$$

**3.20.** **a)** The $s$–domain equivalent circuit:

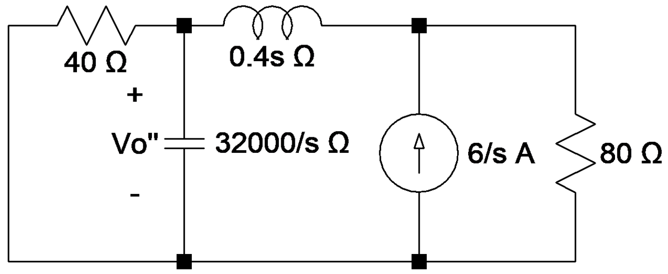



Zero the current source to find the individual response of the voltage source.



$$V_o' = \frac{400}{s} \cdot \frac{\dfrac{(0.4s+80)\cdot \frac{32000}{s}}{0.4s+80+\frac{32000}{s}}}{40 + \dfrac{(0.4s+80)\cdot\frac{32000}{s}}{0.4s+80+\frac{32000}{s}}} = \frac{320000(s+200)}{s(s^2+1000s+240000)}$$

Zero the voltage source.




$$I'' = \frac{80}{\left(\frac{32000}{s} \parallel 40\right) + 0.4s + 80} \cdot \frac{6}{s} = \frac{480(s+800)}{s(0.4s^2+400s+96000)}$$

$$V_o'' = \frac{1200(s+800)}{s(s^2+1000s+240000)} \cdot \left(\frac{32000}{s} \parallel 40\right) = \frac{38400000}{s(s^2+1000s+240000)}$$

$$\Longrightarrow \boxed{V_o = V_o' + V_o'' = \frac{320000(s+320)}{s(s^2+1000s+240000)} \text{ V–s}}$$

**b)** $V_o = \dfrac{K_1}{s} + \dfrac{K_2}{s+400} + \dfrac{K_3}{s+600}$.

$$K_1 = \left.\frac{320000(s+320)}{s^2+1000s+240000}\right|_{s=0} = \frac{1280}{3}, \quad K_2 = \left.\frac{320000(s+320)}{s(s+600)}\right|_{s=-400} = 320$$

$$K_3 = \left.\frac{320000(s+320)}{s(s+400)}\right|_{s=600} = -\frac{2240}{3}$$

$$\Longrightarrow \boxed{V_o(t) = \left(\frac{1280}{3} + 320e^{-400t} - \frac{2240}{3}e^{-600t}\right) u(t) \text{ V}}$$